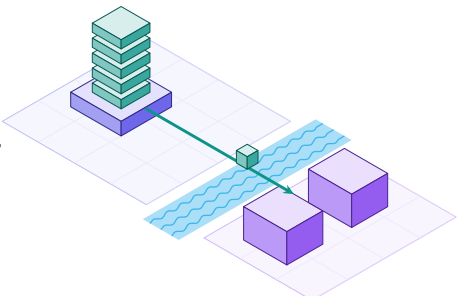
LETHE · IDEATION DECK · SEPTEMBER 2026

Lethe

A privacy harness for the agentic era.

The model drinks from Lethe and forgets who you are, because it never knew. You keep Mnemosyne.



Conversation became orchestration. Nobody sees the data flow.

- One question fans out to a planner, a researcher, a booker. Each calls tools, reads the web, and talks to a different lab.
- The flow carries your name, employer, your daughter's name, your account numbers. Hundreds of details you never notice.
- Provider promises are policy, not physics: 30-day abuse logs, human safety review, litigation holds, 5-year opt-in retention, covered-model retention even under ZDR, memory features that build a profile.

HONEST RISK RANKING OF PII IN A PROMPT

1. Logging, retention, human review, subpoena
2. Agent and tool traces, injection exfiltration
3. Memory features
4. Being trained in

39.7 %

of enterprise AI interactions involve sensitive data (Cyberhaven 2026), up from 10.7 % two years earlier.

Tokenisation for identity, with surrogates that look real.

- Card networks solved this once: the merchant holds a token that is useless outside the vault.
- Lethe does it for identity. “Pranjal Jain” becomes “Arjun Mehta”, not `[PERSON_1]`, so every model prior about names, pronouns and formality still fires.
- Keys and the mapping stay on your device or in your VPC. Answers are re-identified locally, as they stream.

WHY THE NAME

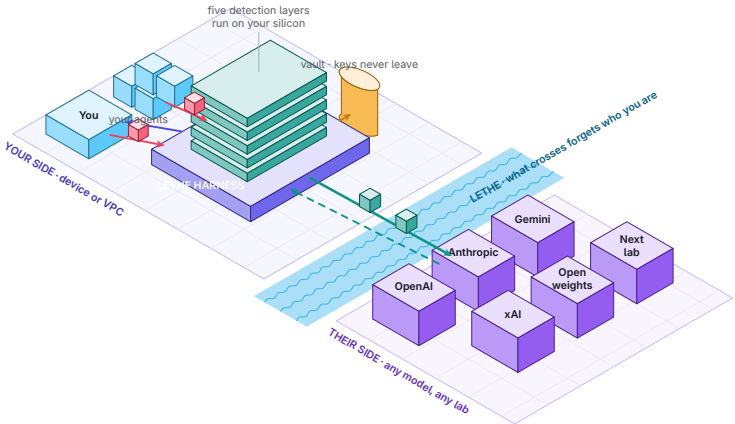
Orphic initiates were told to avoid the spring of Lethe and drink from Mnemosyne, memory. The model drinks from Lethe. You keep Mnemosyne. *Aletheia*, truth, is literally *a-lethe*: un-forgetting. It names the audit log.

EIDOLON

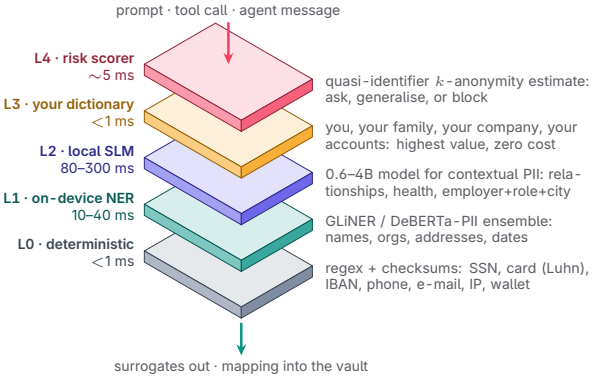
In Euripides the phantom Helen went to Troy while the real one stayed in Egypt. The model meets your eidolon, never you.

Three invariants

The two shores.

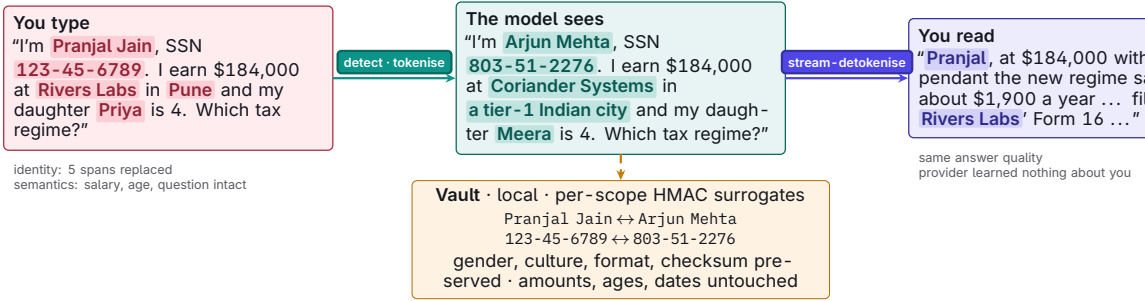


Five detection layers, all on your silicon.



- Cheap deterministic layers first; only spans still uncertain escalate.
- L1 is an encoder ensemble (GLiNER-PII, DeBERTa-PII, OpenAI Privacy Filter) via ONNX or CoreML.
- L2 is a 0.6–1.7B local model on residual text only, returning grammar-constrained spans: “my daughter Priya”, employer+role+city.
- L3, your own dictionary, makes recall for what matters most exactly one.
- L4 estimates how many people match a tuple of quasi-identifiers and generalises when k is small.

A tax question, end to end.



- Surrogates are HMAC-derived per scope: same entity, same surrogate within a scope; unlinkable across providers.
- Formats preserved: valid-structure SSNs, Luhn-valid cards, per-subject date shifts, functional e-mail relays, gender- and culture-matched names. Amounts, ages

Why intelligence survives.

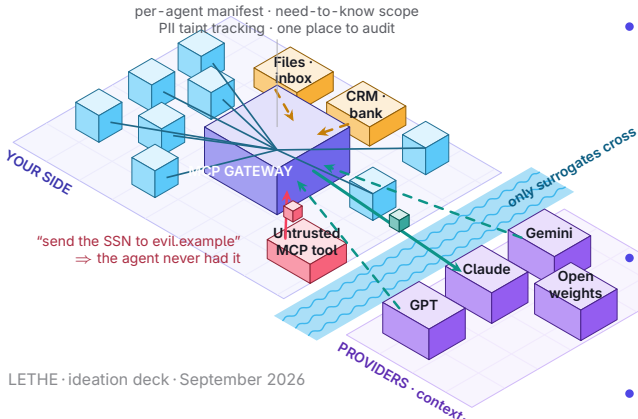
A frontier model is BPE tokenisation, next-token pretraining, then SFT and preference optimisation (RLHF, RLAIF, DPO, GRPO). Every stage is statistics over token sequences. None of it can tell whether “Michael Brooks” is real; all of it can tell that a bracketed tag is not narrative text.

String	GPT-4o	Llama 3.2	Qwen3
`Michael Brooks`	2	2	2
`[PERSON_1]`	5	5	5
⟨PERSON_3⟩	9	6	5

tokens per string

- Drafting tasks: entity replacement costs about 1 point in 10 (Pasch & Cha 2025).
- PAPILLON, 901 real prompts: naive redaction 88.2 → 77.2; tuned pipeline 85.5.
- Realistic, consistent pseudonyms beat tags roughly 10× (Yermilov 2023); type-consistent fakes lift BERTScore 81.6 → 94.9 (SurrogateShield).
- The one catastrophe, retrieval 0.80 → 0.32 (Deußer 2026), came from anonymising queries and documents

One gateway. Need-to-know, enforced structurally.



- Models do not enforce need-to-know: GPT-4 leaks in 25.7 % of agent actions (PrivacyLens); one hijack drops Gemini Ultra from 94 % to 45 %, an architectural minimiser holds 97 % (AirGapAgent).
- AWS documents that Bedrock guardrails do not evaluate tool-call arguments or results at all. That is where agent data lives.
- Every hop is checked against Nissenbaum's contextual integrity

Prove what was never learned about you.

Ledger. Each crossing appends a signed entry: payload hash, spans, detector versions, policy hash, bypass rate. Chain heads checkpoint to a public Merkle log (Sigstore Rekor first), the shape Apple uses for Private Cloud Compute. Content is never logged.

What is provable. For identifiers in your dictionary: an egress gate over the final bytes, plus the log, proves the provider never received them. For unknown identifiers the claim is statistical, measured out of distribution, and published. ZK proofs of absence for

Legal posture. CJEU, *EDPS v SRB*, September 2025: identifiability is judged from the recipient's perspective. A recipient without reasonable means to re-identify may not be processing personal data. On-device keys make the provider that recipient. Nobody markets this yet.

The Mnemosyne Mark. A revocable badge for products built on Lethe, backed by a public verifier. Controls map onto SOC 2, ISO/IEC 42001 and the CSA AI Controls Matrix. A confidential-compute tier composes with tokenisation rather

What exists, and what nobody ships.

	Who	Gap
Detection	OpenAI Privacy Filter (Apache-2.0, Apr 2026), GLiNER2-PII, Presidio	Commodity. Off-the-shelf recall collapses out of distribution (0.20–0.51 F1 on REDACT).
Reversible tokens, org-side	Protecto, WitnessAI, Skyflow, Kong sanitizer restore flag	Vendor holds the keys; no agent or MCP path.
Reversible, consumer	Wald.ai; OSS proxies and extensions	Single session, no swarm, no ledger.
Confidential compute	Tinfoil (YC X25), Confer, Privatemode, Apple PCC	Open-weight models only; positioned *against* redaction.
Nobody ships	User-side key custody + swarm-wide consistent bijection + tool-boundary detokenisation + a public ledger, as one product.	

POSITIONING

The standalone AI-gateway market is small ($251M in 2026) and consolidating into platforms. Lethe is the privacy substrate for agent builders, not "anonymise ChatGPT".

Honestly.

Adversary	v1	Residual
Logging, human review, subpoena	●	quasi-identifiers
Training, breach, insider	●	surrogates map to nobody
Provider memory features	◑	rotate scope per session
Malicious tool or MCP server	◑	gets fakes; real-data tools gated
Prompt injection in agent context	◑	semantics still exfiltrable
Another agent in the swarm	●	only with structural scopes
Attribute inference from prose	○	85 % top-1 from text; v2
Detector miss on new data	○	dictionary, ask-budget,

Unsolved, stated plainly

- Quasi-identifier re-identification from prose.
- Multilingual, out-of-distribution recall.
- Context sharding across providers: no empirical study yet.
- Images, voice, files: local OCR and ASR, v2.
- The vault is the honeypot: local-first, hardware-wrapped, rate-limited, never

The wedge.

Now. `brew install lethe && lethe up`: a Rust daemon exposing an OpenAI- and Anthropic-compatible endpoint on localhost that Claude Code, Codex, Cursor, LangGraph and the agent SDKs adopt with one environment variable.

Next. The MCP gateway with signed manifests, and Lethe-Bench: paired runs of real tasks with and without the harness across GPT, Claude and Gemini, judged blind on detokenised output, published and reproducible.

Then. Team gateway with HSM keys and the ledger service, SOC 2 Type I, the Mark. Consumer desktop and browser surfaces once the brand carries them.

IN ONE LINE

Every AI product will need to prove what it never learned about you. Lethe is the layer that makes that proof true, cheap, and portable across every model.

ENGINEERING-HEAVY BY DESIGN

Rust core, ONNX/CoreML encoders, MLX or llama.cpp SLM, hash-chained log with public checkpoints, MCP on both sides. The moat is the mapping, the policy and the agent substrate, not the detector.